

EXPERIMENT 8

PROBLEM TITLE

Develop a Simple Text Classification Model Using Bag-of-Words.

CODE

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score

# Sample text data
texts = ["I love programming", "Python is amazing", "I dislike bugs",
         "Debugging is fun", "I love solving problems"]
labels = [1, 1, 0, 1, 1] # 1 for positive, 0 for negative

# Step 1: Convert text to Bag-of-Words vectors
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(texts)

# Step 2: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, labels, test_size=0.2,
                                                    random_state=42)

# Step 3: Train a classifier
classifier = MultinomialNB()
classifier.fit(X_train, y_train)

# Step 4: Predict and evaluate
y_pred = classifier.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

EXPECTED OUTPUT

Accuracy: 1.0

EXPERIMENT 9

PROBLEM TITLE

Develop a Simple Text Classification Model Using TF-IDF Representation.

CODE

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

# Sample text data
texts = ["I love programming", "Python is amazing", "I dislike bugs",
        "Debugging is fun", "I love solving problems"]
labels = [1, 1, 0, 1, 1] # 1 for positive, 0 for negative

# Step 1: Convert text to TF-IDF vectors
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(texts)

# Step 2: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, labels, test_size=0.2,
        random_state=42)

# Step 3: Train a classifier
classifier = SVC(kernel='linear')
classifier.fit(X_train, y_train)

# Step 4: Predict and evaluate
y_pred = classifier.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

EXPECTED OUTPUT

Accuracy: 1.0

EXPERIMENT 10

PROBLEM TITLE

Study of PROLOG Programming Language and its Functions.

CODE

% Defining some basic facts

father(john, paul).

father(paul, mark).

mother(mary, paul).

mother(linda, mark).

% Defining a rule to determine if someone is a parent

parent(X, Y) :- father(X, Y).

parent(X, Y) :- mother(X, Y).

% Defining a rule to determine if someone is a grandparent

grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Example Queries:

% ?- parent(john, paul).

% ?- grandparent(john, mark).

% ?- mother(X, mark).

EXPECTED OUTPUT

?- parent(john, paul).

true.

?- grandparent(john, mark).

true.

?- mother(X, mark).

X = linda.